

1 Tsuyo's Innovative Motor Design Revolutionises Conventional Methods

2 Startup Karkhana IO Reinventing The EMS Wheel In India

3 Datoms Revolutionising OEMs With Smart And Connected Products

1 TSUYO'S INNOVATIVE MOTOR DESIGN REVOLUTIONISES CONVENTIONAL METHODS

This startup's motors utilise ferrite magnets instead of rare earth magnets. The aim is to partner with at least 500 car companies over the next five years



TZ Motor Axle Gearbox ASM



AC Induction Motor HT 2



THYHT Controller

Tsuyo Manufacturing, founded in 2020 by three young entrepreneurs (Vijay Kumar, Lalit Baid, and Tripurari Kumar), is a Greater Noida-based EV startup. The founders are serial entrepreneurs and successful professionals from India's top premier institutions like NITs and IIMs/SPJain/IIFT.

The startup manufactures electric motors and controllers for electric vehicles, and they claim to supply them to many OEMs in three-wheeler (low speed and high speed), four-wheeler, and off-road vehicle segments. Within one-and-a-half years of its establishment, the startup has managed to receive funding from Ramkrishna Forgings of \$12 million to support the next five years of expansion and growth.

They have used numerous pat-

terns to design motors. For current supplies, the company is engaged in producing BLDC, PMSM, and AC synchronous electric motors for various applications as per customer end usage and requirement. One notable technology is the permanent magnet-assisted synchronous reluctance motors (PMASynRM), as well as ferrite-based and rare-earth free magnet design. Additionally, SRM-based and axial flux motors are a few new technological frontiers the company is working upon.

"Our designs incorporate new processes and materials, replacing conventional usage of BoM components in the construction of motors like Si-Fe-based motor lamination and new impregnation techniques and winding techniques. These innovations have led to substantial

enhancements in power and performance. Moreover, advancement in drive technology involves the implementation of various algorithms and optimisation techniques, along with controller strategies that improve motor performance within an integrated system," shares Vijay Kumar, Co-Founder of Tsuyo Manufacturing.

This startup also helps in vehicle conversion. "The conversion process extends the vehicle's lifespan, as it can be used beyond the typical 10 to 13 years and reduces emissions by avoiding the purchase of a new vehicle. We are involved in conversion projects for commercial vehicles, three-wheelers (such as those from Bajaj, Tata, and Mahindra), and even diesel forklifts. We also undertake conversion projects for excavators and smaller vehicles like skid steer